Notes: Jensen & Zhang (AER, 2026)
Effects of Parental Death on Labor Market Outcomes

Contents

1 Big picture

- **Question.** What happens to adult children's labor-market outcomes after the death of a parent, and which mechanisms explain the effects?
- **Setting.** The paper uses Danish population-level administrative data from 1980–2019, linking adult children to parents, deaths, causes of death, earnings, labor-market participation, health care use, prescriptions, childcare, and family structure.
- **Treatment.** The main treatment is the first sudden parental death, occurring from 1985 to 2014, for adult children aged 25–50 with two known parents.
- **Identification.** The authors combine the timing of sudden deaths with nearest-neighbor matched controls and a difference-in-differences/event-study design.
- **Main result.** Parental death causes persistent earnings and employment declines. Five years after parental death, men's earnings are about 2 percent lower and women's earnings are about 3 percent lower.
- **Gender pattern.** Mothers' deaths have larger negative earnings effects than fathers' deaths, and daughters experience larger earnings losses than sons.
- **Mental health mechanism.** Both men and women increase psychologist visits and mental-health prescriptions after losing a parent. The timing of mental-health treatment and earnings declines is closely linked.
- **Informal childcare mechanism.** Women with young children suffer especially large earnings declines, close to 4 percent. The paper interprets this as evidence that losing grandparent-provided childcare is important.
- **Eldercare mechanism.** Women whose surviving parent becomes ill experience larger earnings losses, but this channel affects only a smaller subset of adult children.
- **Robustness.** The authors rule out or weaken alternative explanations based on bequests, fertility, cohabitation, job-type switching, relocation, and soon-to-be-treated controls.
- **Economic takeaway.** Parental death is nearly universal, but it is not economically neutral. It creates labor-market losses through grief, mental health, and family-support shocks, with especially important implications for women and mothers of young children.

2 Data and measurement

2.1 Institutional background

- Denmark has universal health care, and general practitioners are the first point of contact for most health issues.
- Hospital and specialist treatment are publicly funded, while psychologist services may involve co-payments but are subsidized for conditions including bereavement after the death of a close relative.
- Formal childcare is heavily subsidized and widely used, but grandparents still provide important informal childcare, especially for after-school gaps, sick children, and working parents' scheduling constraints.
- Eldercare is publicly provided by municipalities, but some informal eldercare is still provided by family members.
- Inheritance after the first parental death is often limited because the surviving spouse can often remain in an undivided estate.

Interpretation: The institutional background helps interpret mechanisms. Denmark already has strong formal health care, childcare, and eldercare systems, so observed earnings losses are not simply due to absent public services. If effects are visible in Denmark, they may be stronger in settings with weaker public support.

2.2 Population registers

- The analysis uses Danish register data that observe parent-child links, demographics, death timing, and cause of death.
- Labor-market outcomes include employment, earnings, and a proxy for hours worked based on ATP pension contributions.
- Earnings include income from employment and self-employment, are inflation-adjusted, and are winsorized to limit extreme outliers.
- Health outcomes include GP, psychologist, psychiatrist, and hospital visits, plus prescription records.
- Prescription outcomes include mental-health prescriptions and opioid painkiller prescriptions.
- Childcare registers identify formal childcare enrollment at the child level from 1995 onward.

Interpretation: The paper's key advantage is that the data cover the entire population and many channels simultaneously: labor supply, earnings, health care, mental health, childcare, and family structure.

2.3 Treatment group

- The treatment sample consists of adult children whose first parental death was sudden and unexpected.
- Sudden causes include heart disease, cerebrovascular disease, acute respiratory infections, and traffic or external accidents.
- The paper focuses on first parental deaths to reduce confounding from bequests and eldercare for the first deceased parent.
- The treatment window is 1985–2014 so that the authors can observe pre- and post-treatment outcomes.
- Adult children are restricted to ages 25–50 at the time of first parental death and must have two known parents.

Interpretation: Sudden first parental deaths help address two threats: anticipation of a parent's death and reverse causality from the child's economic distress to the parent's health.

2.4 Summary statistics

- Table 1 compares children whose first parental death is sudden with those whose first parental death is nonsudden.
- The sudden-death group has 206,286 observations; the nonsudden-death group has 528,057 observations.
- Adult children in both groups are very similar: average age is about 37, male share is about 0.52–0.53, employment is 0.84 in both groups, and share with children under six is 0.34 in both groups.
- Parents who die suddenly are slightly older, but the differences in adult-child characteristics are modest.

Interpretation: The comparison supports external validity: the sudden-death sample does not appear wildly different from the broader population experiencing parental death.

3 Empirical specifications and interpretation

3.1 Descriptive event study

Before introducing matched controls, the paper estimates a descriptive event-study model within the treated sample:

$$Y_{ity} = \sum_{k=-5,\, k\neq -1}^{5} \delta_k \mathbf{1}\{k=t\} + AgeYear_{iy} + EduYear_{iy} + \varepsilon_{ity}. \tag{1}$$

- Y_{ity} is the outcome for individual i in calendar year y, where t is event time relative to first parental death.
- Year $t = -1$ is omitted, so coefficients are interpreted relative to the year before parental death.
- Age-by-year and education-by-year controls allow age and education effects to vary over the long sample period.

Interpretation: This specification shows raw dynamics around parental death, but it does not fully solve concerns about counterfactual trends. It motivates the matched-control DiD design.

3.2 Nearest-neighbor matching

The main design matches each treated individual to a control individual who does not experience parental death within the treated individual's five-year post-event window.

- Exact matching variables: gender, earnings group, and whether the youngest child is under age six.
- Mahalanobis nearest-neighbor variables: age, education, public/private employment sector, cohabitation, residential region, number of children, parental ages, age of youngest child, and three-year histories of employment and earnings.
- Matching is done one year before parental death.

Interpretation: The matching is designed to make treated and control adults comparable in demographics, family structure, and pre-event labor-market trajectories.

3.3 Matched event-study specification

Using treated individuals and matched controls, the paper estimates:

$$Y_{ity} = \sum_{k=-5,\, k\neq -1}^{5} \left(\delta_k \mathbf{1}\{k=t\} D_i + \theta_k \mathbf{1}\{k=t\}\right) + \gamma_i + \gamma_y + Age_{iy} + \varepsilon_{ity}. \tag{2}$$

- $D_i = 1$ for treated individuals and 0 for matched controls.
- δ_k measures the treatment-control difference at event time k, relative to year -1.
- γ_i are individual-by-match fixed effects.
- γ_y are calendar-year fixed effects.
- θ_k are event-time fixed effects common to treated and control individuals.
- Standard errors are clustered at the individual-by-match level.

Interpretation: The event study is the main visual test for pre-trends and treatment dynamics. If treated and control outcomes move together before year 0, the matched control group is plausible.

3.4 Average DiD specification

For aggregate post-treatment effects, the paper estimates:

$$Y_{ity} = \beta D_i \times post_t + \gamma_i + \gamma_t + \gamma_y + Age_{iy} + \varepsilon_{ity}. \tag{3}$$

- $post_t = 1$ after parental death.
- β is the average treatment effect over the post-death window relative to matched controls.
- The model is estimated separately for men and women.

Interpretation: This coefficient summarizes the average labor-market effect of parental death across the post-treatment years.

3.5 Mother versus father specification

To compare deaths of mothers and fathers, the paper estimates:

$$Y_{ity} = \beta_m M_i \times post_t + \beta_f F_i \times post_t + \gamma_i + \gamma_t + \gamma_y + Age_{iy} + \varepsilon_{ity}, \tag{4}$$

where $M_i = 1$ if the first deceased parent is the mother and $F_i = 1$ if the first deceased parent is the father.

Interpretation: The two coefficients identify whether maternal and paternal deaths have different labor-market and health consequences, separately for sons and daughters.

4 Identification and empirical design

4.1 What identifies the main effect

- The design compares treated individuals with matched controls who look similar before the event.
- Treatment timing is based on sudden first parental death.
- The matched event study uses within-pair comparisons around the event.
- Pre-event coefficients provide evidence on whether treated and controls were moving similarly before parental death.

Interpretation: The identifying assumption is conditional parallel trends: absent the first sudden parental death, treated individuals would have followed the same outcome path as their matched controls.

4.2 Why sudden first deaths matter

- Sudden deaths reduce anticipation: adult children are less likely to adjust labor supply before the death.
- Sudden deaths reduce reverse causality: children's employment or earnings shocks are less likely to cause the parent's sudden death.
- First deaths reduce bequest concerns because major inheritance events are more relevant after the second parental death.
- First deaths reduce eldercare concerns for the deceased parent because the spouse often provides primary care before death.

Interpretation: The sample restriction is both an identification strategy and a way to isolate specific mechanisms: mental health and family support rather than bequests or long pre-death caregiving.

4.3 Design strengths

- Population-level data allow large samples and fine gender interactions.
- Register data avoid attrition and survey recall bias.
- Matched controls are not already-treated individuals, avoiding the main "forbidden comparison" issue in staggered DiD settings.
- Rich outcomes allow the authors to trace mechanisms, not only final earnings effects.

4.4 Remaining limits

- Sudden deaths are not literally randomly assigned.
- Matching cannot eliminate all unobserved family-level differences.
- Some health outcomes are available only after 1990 or 1995, limiting mechanism samples.
- The exact magnitude of effects may differ for nonsudden parental deaths or in countries with different childcare, health care, or eldercare systems.
- Mechanisms can overlap: mental health, childcare, and surviving-parent health may interact.

Interpretation: The paper's evidence is strongest for the causal effect of sudden first parental death in Denmark. Mechanism estimates are persuasive but not a full structural decomposition.

5 Tables and figures

This section keeps the same compact format: each visual block includes **Read** plus **Economic message**, with adjacent original-paper visuals grouped to save space.

5.1 Table 1 and Figure 1: sample balance and main dynamic effects

Read:

- Table 1 shows that the sudden-death and nonsudden-death groups are broadly similar in adult-child characteristics.
- Employment is 0.84 in both groups; the share with children under six is 0.34 in both groups.
- The sudden-death group has 206,286 observations, and the nonsudden-death group has 528,057 observations.
- Figure 1 shows little evidence of pre-trends before parental death.
- After parental death, employment, hours proxy, raw earnings, and indexed earnings all decline.
- Five years after parental death, men's earnings are roughly 2 percent lower and women's earnings are roughly 3 percent lower.

Economic message: The treatment sample is large and comparable to the broader parental-death population, while the main event study shows persistent labor-market losses after parental death.

TABLE 1—SUMMARY STATISTICS FOR ADULT CHILDREN AT $\tau = -1$

	Sudden death mean	Nonsudden death mean	p-value p
Age	36.98	36.88	0.00
Male	0.53	0.52	0.00
Share with college or above	0.21	0.22	0.00
Share with high school	0.51	0.52	0.00
Share without high school	0.29	0.26	0.00
Cohabitation	0.71	0.70	0.00
Number of children	1.39	1.38	0.02
Age of youngest child	7.94	7.76	0.00
Share with children under 6	0.34	0.34	0.00
Mother age	65.44	64.28	0.00
Father age	69.57	67.93	0.00
Mother married	0.85	0.81	0.00
Father married	0.85	0.82	0.00
First death age	70.00	68.00	0.00
Employment	0.84	0.84	0.31
Intensive margin	1,016.34	1,020.04	0.00
Annual earnings	296.80	303.33	0.00
Observations	206,286	528,057	734,343

Notes: This table shows the summary statistics for children in the calendar year preceding

(a) Table 1: summary statistics

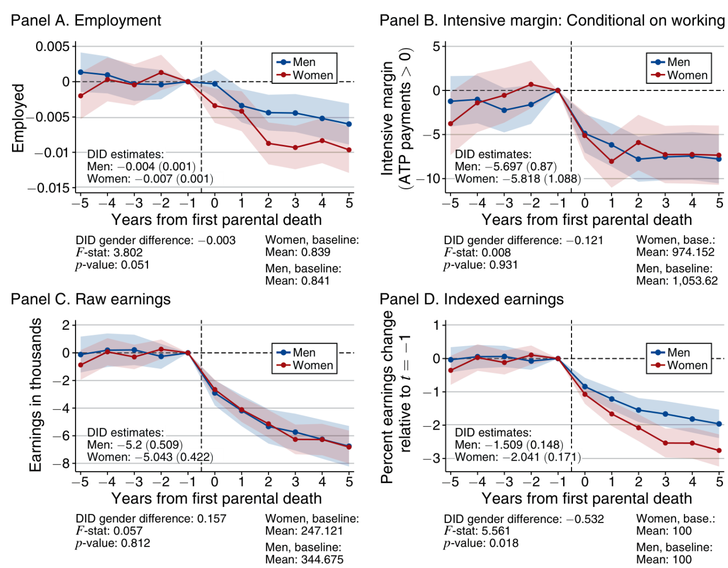


FIGURE 1. EFFECT OF PARENTAL DEATH BY GENDER

(b) Figure 1: employment and earnings dynamics

5.2 Figure 2 and Figure 3: gender interactions and health mechanisms

Read:

- Figure 2 shows that mothers' deaths generate larger earnings reductions than fathers' deaths for both men and women.
- Women experience larger earnings penalties than men after both maternal and paternal deaths.

- Figure 3 shows that parental death increases psychologist visits and mental-health prescriptions for both genders.
- Mothers' deaths generate especially large increases in psychologist visits and mental-health prescriptions.
- Opioid prescriptions and hospital visits also rise after parental death, although the strongest evidence is for mental health.

Economic message: The labor-market shock is gendered in two ways: mothers matter more than fathers for adult-child earnings, and daughters experience larger earnings losses. Mental health is a major mechanism connecting bereavement to labor-market outcomes.

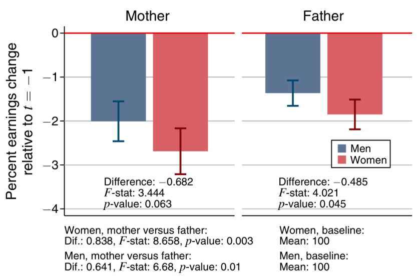
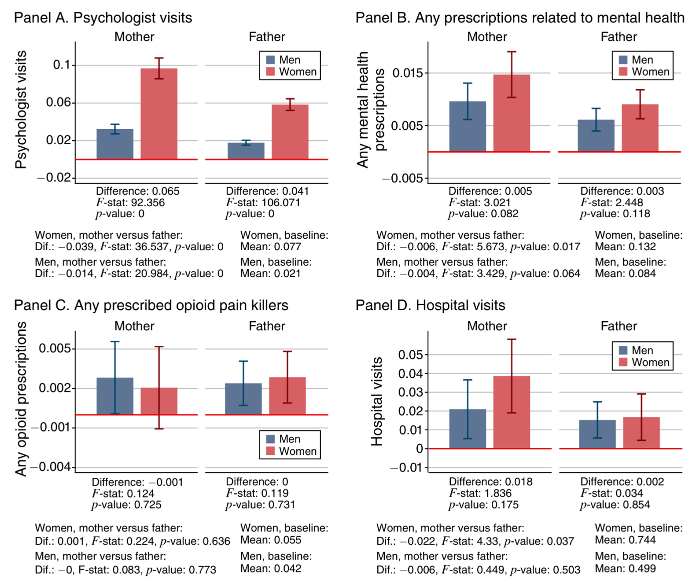


FIGURE 2. EFFECTS BY PARENTAL AND CHILD GENDER INTERACTIONS

(a) Figure 2: parent-child gender effects



(b) Figure 3: mental-health and health outcomes

5.3 Figure 4 and Figure 5: timing of mental health and informal childcare

Read:

- Figure 4 shows a spike in year-to-year earnings declines immediately after parental death.
- The start of mental-health prescriptions also rises at the time of parental death.
- The co-occurrence panel shows that mental-health treatment and earnings declines are linked at the event.
- Figure 5 separates adults by whether they have young children.
- Women with young children experience the largest earnings decline, close to 4 percent.
- Men with young children are less affected, and adults without young children have smaller losses.

Economic message: The mental-health channel is visible in timing, while the childcare channel is visible in heterogeneity: the loss of grandparents' informal childcare falls especially on women with young children.

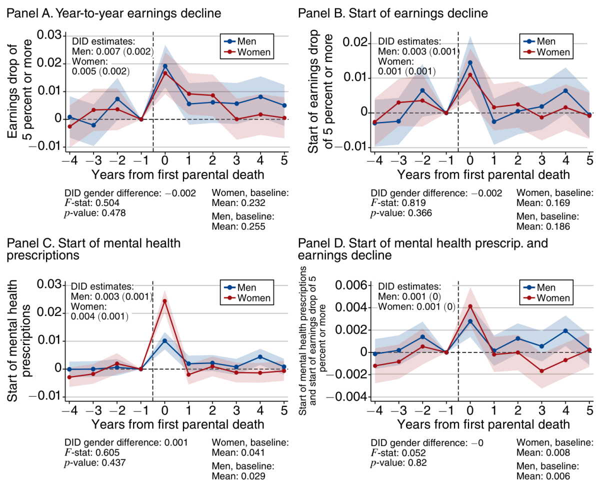


FIGURE 4. CO-OCCURRENCE OF EARNINGS DECLINES AND MENTAL HEALTH TREATMENT START

(a) Figure 4: earnings declines and mental-health treatment

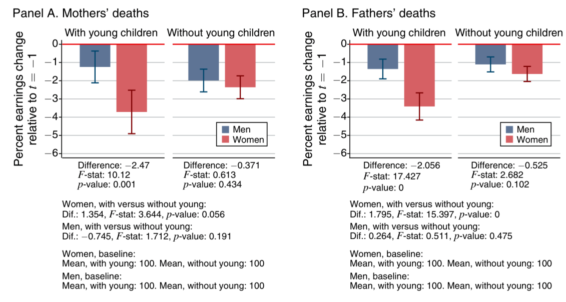


FIGURE 5. EARNINGS BY YOUNG CHILDREN, MOTHERS' VERSUS FATHERS' DEATHS

(b) Figure 5: earnings by young-child status

5.4 Figure 6: surviving parent health and eldercare

Read:

- Figure 6 restricts the sample to adults without young children and splits by whether the surviving parent is ill.
- Women experience larger earnings losses when the surviving parent is ill.
- The ill-versus-not-ill difference for women is about 0.95 percentage points and statistically significant.
- For men, the difference is smaller and not as precisely estimated.
- The paper argues that this is suggestive evidence of an eldercare channel, but not the dominant driver of the overall earnings loss.

Economic message: Eldercare can matter, especially for women, but the main population-level mechanisms are mental health and informal childcare because most surviving parents remain relatively healthy.

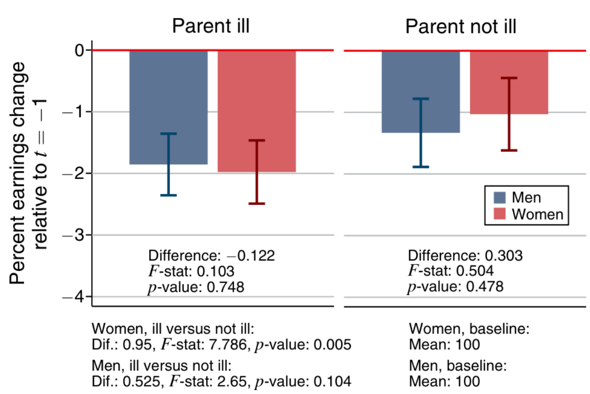


FIGURE 6. EARNINGS BY SURVIVING PARENT'S HEALTH STATUS

Figure 4: Figure 6: earnings by surviving parent's health status

6 Short summary

- The paper studies the effects of first parental death on adult children's employment, earnings, mental health, childcare, and family-support outcomes.
- It uses Danish population registers from 1980–2019 and first sudden parental deaths from 1985–2014.
- The empirical design combines sudden death timing, nearest-neighbor matched controls, event studies, and DiD regressions.
- Adult children suffer persistent employment and earnings declines after parental death.
- Men's earnings decline by about 2 percent and women's earnings by about 3 percent five years after parental death.
- Mothers' deaths have larger negative effects than fathers' deaths.
- Daughters experience larger earnings declines than sons.
- Both genders increase psychologist visits and mental-health prescriptions after parental death.
- Earnings declines and mental-health treatment starts co-occur around the death event.
- Women with young children experience especially large earnings losses, close to 4 percent, consistent with loss of informal childcare.
- Women whose surviving parent is ill also experience larger losses, suggesting an eldercare channel.
- Robustness checks weaken alternative mechanisms such as bequests, fertility, cohabitation, job-type changes, and relocation.
- The core policy implication is that bereavement support is economically relevant, not only emotionally relevant.

7 Conclusion

7.1 Core conclusion

Jensen and Zhang show that parental death has persistent effects on adult children's labor-market outcomes. The event reduces earnings and employment for both men and women, with larger earnings losses for women and larger effects after mothers' deaths. The effects persist beyond the immediate shock period, making parental death a meaningful labor-market event as well as a family and health event.

7.2 Economic intuition

Parental death changes adult children's constraints and well-being. Grief and mental-health deterioration can reduce labor supply and productivity. The death of a grandparent can also remove informal childcare, especially harming women with young children. If the surviving parent becomes ill, adult children may additionally face eldercare obligations. These channels convert a family shock into an earnings shock.

7.3 Takeaway

The paper reframes bereavement as an economic shock with gendered labor-market consequences. Even in Denmark, where formal health care, childcare, and eldercare are extensive, parental death lowers adult children's earnings. The results suggest that grief support, mental-health access, and family-care policies can have labor-market relevance.